Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 2026
Team ID	LTVIP2026TMIDS36686
Project Name	Project – Unveiling sales impact with tableau visualization

Technical Architecture:

The technical architecture of Unveiling Sales Impact with Tableau Visualization is designed using a structured three-tier architecture model to ensure scalability, performance, security, and real-time analytics. The architecture consists of the Data Layer, Application Layer, and Presentation Layer, integrated with cloud infrastructure and external mode

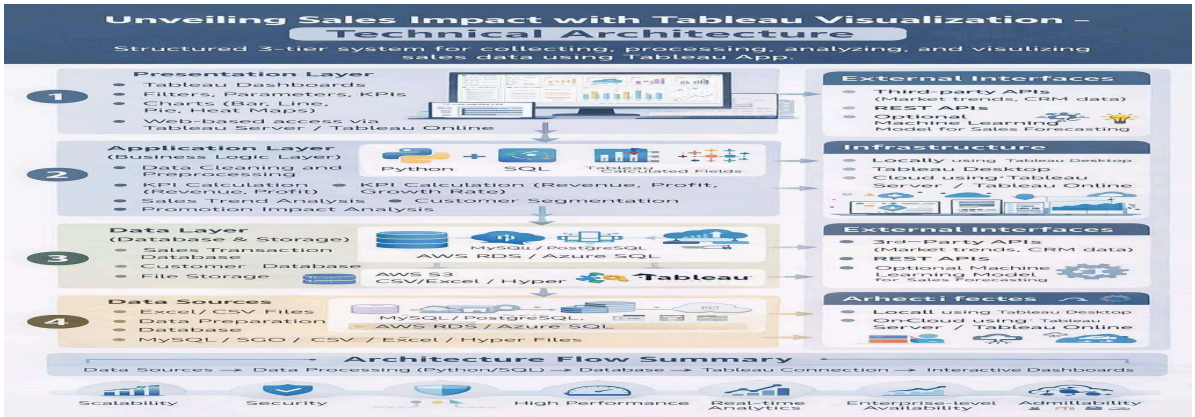

Table-1 : Components & Technologies:

Si no	Component	Description	Technology
1.	User Interface	How users interact with the application through interactive dashboards, filters, charts, and reports.	tableau Desktop, Tableau Public, Tableau Server, Web Browser (HTML, CSS – frontend view)
2.	Application Logic – 1 (Data Collection Layer)	Logic for collecting sales data from different sources such as Excel files, CSV files, or databases.	Python, Pandas, SQL, Excel Connector, Tableau Data Connector
3.	3. Application Logic – 2 (Data Processing Layer)	Logic for data cleaning, transformation, aggregation, and KPI calculation (Revenue, Profit, Growth Rate).	Python (Pandas, NumPy), SQL Queries, Tableau Calculated Fields
4.	4. Application Logic – 3 (Visualization & Analysis Layer)	Logic for generating dashboards, charts, filters, and applying business rules for insights	Tableau Desktop (Charts, Parameters, Filters, LOD Expressions)
5.	database	Stores structured sales data including customer, product, and transaction details.	MySQL / PostgreSQL / SQL Server
6.	Cloud Database	Cloud-based database service for scalable and remote access to sales data.	. AWS RDS / Google Big Query / Azure SQL Database
7.	File Storage	Description: Stores raw datasets, extracted files (. hyper), and report exports (PDF, Excel).	AWS S3 / Google Cloud Storage / Local Storage
8.	External API-1	API used for importing additional datasets like market trends or competitor pricing.	REST API, JSON Integration.

9.	External API-2	API for integrating customer demographics or thirdparty analytics tools.	. Technology: Public Data API / CRM API Integration
10.	10. Machine Learning Model (Optional – if applicable)	Used for sales forecasting, trend prediction, and customer segmentation	Python (Scikit-learn), Time Series Forecasting Model
11.	Infrastructure (Server / Cloud)	Application deployment environment (Local system or Cloud platform).	Tableau Server / Tableau Online AWS / Azure / Google Cloud Load Balancer & Backup System:

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Utilized for data cleaning, preprocessing, and ETL (Extract, Transform, Load) processes before visualization.	Python (Pandas, NumPy), Scikit-Learn
2.	Security Implementations	Role-based access control (RBAC) to ensure only authorized stakeholders view specific sales regions; data encryption at rest.	Tableau Server Permissions, SSL/TLS, OAuth 2.0
3.	Scalable Architecture	A modular approach where data storage is separate from the visualization layer, allowing for massive datasets.	Snowflake or AWS S3 (Cloud Data Warehouse), Tableau Hyper Engine
4.	Availability	High availability via cloud hosting, ensuring the sales dashboard is accessible 24/7 for global teams.	Tableau Cloud / AWS Multi-AZ Deployment
5.	Performance	Use of data extracts and indexed queries to ensure dashboards load in under 2 seconds despite millions of rows.	Tableau Extracts (. hyper), Redis (Caching), CloudFront CDN